

Assignment 2 - E-Commerce Sales Analysis

Financial figures use net sales after discounts, after exact duplicate removal and exclusion of nonpositive quantities.

Q1. Load the dataset.

Answer:

```
df = pd.read_csv("ecommerce_orders_raw.csv")
```

Q2. Display its structure and dimensions.

Answer:

Raw shape: **15,100 rows x 11 columns**. Use `df.info()` and `df.shape`.

Q3. Identify the data types of all columns.

Answer:

Order_ID, Customer_ID, City, Product_Category, Product, Payment_Method, Order_Date: object/string; Quantity and Discount: integer; Unit_Price and Rating: floating point. Parse Order_Date to datetime before analysis.

Q4. Identify missing values.

Answer:

Rating has **401 missing values** (2.66%); all other columns have none.

Q5. Identify duplicate orders.

Answer:

There are **100 exact duplicate rows** (`df.duplicated().sum()`).

Q6. Examine unique values in City, Product Category, and Payment Method.

Answer:

Raw City has 10 spellings because of case/space variants; standardized City has 8: Bangalore, Chennai, Coimbatore, Delhi, Hyderabad, Kolkata, Mumbai, Pune. Product_Category has 6 values: Beauty, Books, Clothing, Electronics, Home Appliances, Sports. Payment_Method has 5 values: Cash on Delivery, Credit Card, Debit Card, Net Banking, UPI.

Q7. Handle missing ratings.

Answer:

Ratings are subjective and cannot be inferred reliably. Preserve missing values and label them "Unrated" in Rating_Category; exclude them from average-rating calculations.

Q8. Standardize city names.

Answer:

```
df["City"] = df["City"].str.strip().str.title()
This combines " chennai ", "CHENNAI", and "Chennai".
```

Q9. Identify invalid quantities.

Answer:

There are **20 rows with Quantity <= 0**. These cannot represent a valid sale and are excluded from revenue analysis.

Q10. Handle duplicate orders.

Answer:

Remove exact duplicate rows with `df = df.drop_duplicates()`. This removes 100 rows. Do not deduplicate only on Customer_ID because customers can place multiple orders.

Q11. Convert Order_Date into an appropriate datetime format.

Answer:

```
df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
All dates parse successfully (0 invalid dates).
```

Q12. Create Gross_Amount = Quantity x Unit_Price.

Answer:

```
df["Gross_Amount"] = df["Quantity"] * df["Unit_Price"]
```

Q13. Create Discount_Amount = Gross_Amount x Discount / 100.

Answer:

```
df["Discount_Amount"] = df["Gross_Amount"] * df["Discount"] / 100
```

Q14. Create Net_Amount = Gross_Amount - Discount_Amount.

Answer:

```
df["Net_Amount"] = df["Gross_Amount"] - df["Discount_Amount"]  
Use Net_Amount for revenue results below.
```

Q15. Create Rating_Category: 1-2 Poor; 3 Average; 4 Good; 5 Excellent.

Answer:

```
rating_map = {1: "Poor", 2: "Poor", 3: "Average", 4: "Good", 5: "Excellent"}  
df["Rating_Category"] = df["Rating"].map(rating_map).fillna("Unrated")
```

Q16. Extract Year, Month, Day, and Day of Week from Order_Date.

Answer:

```
df["Year"] = df.Order_Date.dt.year  
df["Month"] = df.Order_Date.dt.month  
df["Day"] = df.Order_Date.dt.day  
df["Day_of_Week"] = df.Order_Date.dt.day_name()
```

Q17. Calculate total revenue.

Answer:

After the cleaning rules above, total Net_Amount = **\$1,981,274,066.40**.

Q18. Calculate the total number of orders.

Answer:

14,980 unique orders after deduplicating and excluding nonpositive-quantity orders.

Q19. Find the average order value.

Answer:

Mean net order value = total Net_Amount / unique Order_ID = **\$132,261.29**.

Q20. Find total revenue by product category.

Answer:

Beauty \$330,989,843.70; Books \$332,776,839.15; Clothing \$345,417,322.64; Electronics \$323,517,815.01; Home Appliances \$318,936,301.54; Sports \$329,635,944.36. **Clothing** is highest.

Q21. Find the top 10 products by revenue.

Answer:

Jeans \$93,528,952.02; Jacket \$88,841,770.48; Self Help \$88,341,881.88; Yoga Mat \$88,323,335.28; Shampoo \$88,223,909.53; Skin Serum \$87,182,348.17; Laptop \$86,973,276.68; Mixer \$85,292,692.36; Headphones \$83,583,034.67; Textbook \$83,477,808.65.

Q22. Find the city generating the highest revenue.

Answer:

Chennai, \$285,348,511.77.

Q23. Find the most frequently used payment method.

Answer:

Debit Card and Cash on Delivery tie at **3,056 orders each**.

Q24. Find the average rating for each product category.

Answer:

Beauty 3.89; Books 3.85; Clothing 3.91; Electronics 3.83; Home Appliances 3.85; Sports 3.88. Missing ratings are excluded.

Q25. Calculate monthly revenue.

Answer:

Net revenue by month: 2025-01 \$107,940,613.64; 2025-02 \$96,852,305.33; 2025-03 \$105,875,846.25; 2025-04 \$107,323,581.74; 2025-05 \$106,394,813.70; 2025-06 \$102,402,049.82; 2025-07 \$107,413,187.56; 2025-08 \$107,256,978.34; 2025-09 \$108,338,403.04; 2025-10 \$108,666,902.45; 2025-11 \$104,974,427.76; 2025-12 \$103,494,318.10; 2026-01 \$110,837,307.04; 2026-02 \$94,770,074.27; 2026-03 \$101,448,554.16; 2026-04 \$99,666,356.60; 2026-05 \$106,427,959.40; 2026-06 \$99,417,391.05; 2026-07 \$101,772,996.13.

Q26. Identify the highest-revenue month.

Answer:

January 2026, \$110,837,307.04.

Q27. Find the top 10 customers by total spending.

Answer:

Customer ID and spend: C01527 \$2,398,927.85; C01856 \$2,263,955.71; C02170 \$2,179,011.48; C01626 \$2,177,251.92; C01882 \$2,104,829.42; C00990 \$2,057,788.90; C02988 \$2,009,290.05; C02588 \$2,002,666.45; C00139 \$1,988,372.56; C00944 \$1,971,526.54.

Q28. Compare revenue generated by different cities.

Answer:

Bangalore \$239,769,696.33; Chennai \$285,348,511.77; Coimbatore \$246,279,362.92; Delhi \$246,260,476.88; Hyderabad \$232,697,633.02; Kolkata \$239,270,849.20; Mumbai \$247,814,519.36; Pune \$243,833,016.92.

Q29. Find which product category has the highest average order value.

Answer:

Average net order value within each category: Beauty \$131,763.47; Books \$132,001.92; Clothing \$135,351.62; Electronics \$130,872.90; Home Appliances \$130,231.24; Sports \$133,240.07. **Clothing** is highest.